UNSUPERVISED LEARNING | SINGULAR VALUE DECOMPOSITION ©

Conceptual & Theory-Based Questions

1. What is Singular Value Decomposition (SVD)?

📘 **Mathematical Explanation:**

Singular Value Decomposition (SVD) is a fundamental matrix factorization technique. For any real (or complex) matrix **A** of size m × n, SVD decomposes it into three matrices:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

Where:

- **U** is an m × m orthogonal matrix (left singular vectors),
- **Σ (Sigma)** is an m × n diagonal matrix with non-negative real numbers on the diagonal (singular values),
- **V^t** is the transpose of an n × n orthogonal matrix (right singular vectors).

📌 **Key Properties:**

- The singular values (diagonal elements of Σ) are always ≥ 0 and are sorted in descending order.
- The number of non-zero singular values equals the **rank** of matrix A.
- The columns of U are eigenvectors of AA^T, and columns of V are eigenvectors of A^TA.

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🧠 **Intuitive Understanding:**

Imagine the matrix **A** as a transformation that stretches, rotates, or flips space. SVD breaks this transformation into **three simpler steps**:

1. **Rotate/flip the input space (V^t)**
    - Reorients the input basis vectors.
2. **Scale/stretch along new axes (Σ)**
    - Applies a stretch/shrink operation along the new axes (like zooming in or out).

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3. **Rotate/flip again (U)**
    - Transforms the output back into the final space.

So, **SVD tells you how a matrix reshapes space** — first by reorienting it, then scaling it, then reorienting again.

🎯 **Why It's Useful:**

- Works for **any matrix** (square or rectangular).
- Core to **PCA**, **image compression**, **latent semantic analysis (NLP)**, and **recommendation systems**.
- Helps in **dimensionality reduction** by keeping only the largest singular values (truncated SVD).

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2. How is SVD related to PCA (Principal Component Analysis)?

📘 **Short Answer:**

SVD is the mathematical engine behind PCA. While PCA is a statistical method for identifying directions of maximum variance in data, **SVD is the most stable and efficient way to compute it**.

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🔍 Under the Hood: How SVD Computes PCA

Let's break it down:

1. Start with your data matrix X:

- Shape: $m \times n$ (m samples, n features)
- Assume the data is centered (mean of each column = 0)

2. PCA: Find the directions (components) that explain the most variance

- These are the eigenvectors of the covariance matrix:

$$\text{Cov}(X) = \frac{1}{m-1} X^T X$$

3. SVD sidesteps the covariance matrix

Instead of computing $X^T X$, we apply:

$$X = U \Sigma V^T$$

Where:

- **U:** left singular vectors (basis for samples),
- **Σ:** singular values (related to variance),
- **V:** right singular vectors = **principal components** (basis for features)

🔑 Key Insight:

- The **columns of V** (right singular vectors) are the **principal directions** — the axes along which the data varies the most.
- The **singular values** (Σ) squared and divided by m−1m - 1 give the **explained variance** for each principal component.

$$\text{Explained variance}_i = \frac{\sigma_i^2}{m-1}$$

📊 Why Use SVD Instead of Eigen-decomposition?

- SVD is **numerically more stable**, especially when the feature matrix is **not square** or **ill-conditioned**.
- Works even when (X^T).X is **not full rank** or **singular**.

🧠 **Summary Table:**

PCA Concept	SVD Equivalent
Principal components	Columns of V
Explained variance	Squared singular values of Σ
PCA transformation (scores)	UΣ

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3. What are the components of SVD? What do U, Σ, and V^t represent?

🔍 **What is SVD?**

SVD stands for **Singular Value Decomposition**. It's a matrix factorization technique used in linear algebra, machine learning, and data compression.

Given a matrix AA of size m×nm \times n, SVD factorizes it into three matrices:

$$A=U\,\Sigma\,V^T$$

🧱 **Components of SVD:**

Component	Size	Meaning
U	m×m	Left singular vectors (columns are orthonormal)
Σ (Sigma)	m×n	Diagonal matrix with singular values (non-negative)
V^t (V Transpose)	n×n	Right singular vectors (rows of V are orthonormal)

🔵 **U (Left Singular Vectors)**

- Each column is a **basis vector** for the original data in row space.
- Represents the **directions of variance** in the original feature space.
- Think of them like **"eigenvectors"** of AA^T.

🟡 **Σ (Sigma - Singular Values)**

- Contains **sorted singular values** (from largest to smallest) on the diagonal.

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- Represents the **strength/importance** of each corresponding dimension.
- Square root of eigenvalues from $\mathbf{A^T A}$ or $\mathbf{AA^T}$.

🔴 **V^t (Transpose of Right Singular Vectors)**

- Rows represent **directions in feature space** (columns of original matrix).
- V is like the eigenvectors of $\mathbf{A^T A}$.
- Tells us how to **reconstruct** the original data from the compressed version.

💡 **Intuition:**

- **U** gives the orientation of the data (rows).
- **Σ** scales the data (how much variance in each direction).
- **V^t** gives the orientation of the features (columns).

🧪 **Example:**

Say you have a data matrix of images, documents, or user ratings.

SVD helps with:

- **Dimensionality reduction** (keep top-k singular values)
- **Noise reduction**
- **Latent semantic analysis** (e.g., topic extraction from text)
- **Recommender systems** (e.g., Netflix, YouTube)

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4. What are singular values? What do they represent geometrically or statistically?

🔍 **What Are Singular Values?**

Singular values are the **diagonal elements** in the Σ matrix from the **Singular Value Decomposition (SVD)**:

$A = U \Sigma V^T A$

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They are always **non-negative real numbers**, typically ordered from **largest to smallest**.

📊 Statistically Speaking:

Singular values represent the **importance** or **energy** (variance) captured by each **latent dimension** (like principal components in PCA).

If you square each singular value σ_i, you get the **eigenvalues** of the matrix $A^T A$ or AA^T.

So in short:

Larger singular values = more variance / more information
Smaller ones = less important, possibly noise

📐 Geometrically Speaking:

Imagine applying a matrix transformation to a circle in space.

- The **singular values** tell you how much that transformation **stretches or compresses** the shape **along certain directions** (given by U and V).

In 2D:

- A unit circle becomes an ellipse.
- The lengths of the ellipse's axes = **singular values**.

📌 Quick Summary:

Term	Interpretation
Singular Values	Measure of the strength/importance of each axis
Large Value	Captures major pattern/variance
Small Value	Often corresponds to noise or minor variance

🧪 In Practice:

In ML and data science, we often:

- **Keep only the top-k singular values** (for dimensionality reduction)

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- Use them to estimate **rank** of a matrix
- Remove **noise** by discarding small singular values

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5. What's the difference between eigenvalues/eigenvectors and singular values/vectors?

🔁 **Eigenvalues & Eigenvectors vs. Singular Values & Vectors**

Concept	**Eigenvalues & Eigenvectors**	**Singular Values & Vectors (SVD)**
Applies To	**Square matrices only** (n × n)	**Any m × n matrix** (square or rectangular)
Equation	$Av=\lambda v$	$A=U\Sigma V^T$
Output	- Eigenvalues λ Eigenvectors v	- Singular values σ\sigma- Left vectors (U), Right vectors (V)
Matrix type	Diagonalizable (if possible)	Always works! (SVD always exists)
Geometric meaning	Stretching along a direction	Stretching + rotating in multiple directions

🔷 **Eigenvalues & Eigenvectors:**

- Only defined for **square matrices**.
- **Eigenvector**: a direction that remains unchanged (just scaled).
- **Eigenvalue**: how much that direction is scaled.

📌 Used in:

- PCA (via covariance matrix)
- Graph analysis
- Differential equations

🔶 **Singular Values & Vectors (SVD):**

- Defined for **any matrix** — square or rectangular.

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- Always exists, even if eigenvalues don't.
- More general and robust than eigen-decomposition.

📌 Used in:

- Dimensionality reduction
- Recommender systems
- Image compression
- NLP (e.g., Latent Semantic Analysis)

🔁 **Relationship Between Them:**

If A is **square and symmetric**, then:

- **Singular values** of A = **absolute values of eigenvalues**
- **Left and right singular vectors** = **eigenvectors of A**

If A is **non-square**, SVD still works — but eigen-decomposition does not.

🎯 **Final Takeaway:**

Topic	**Eigen**	**SVD**
Only for square?	✅	❌ (any shape)
Always exists?	❌	✅
Practical usage	Math-heavy	Widely used in ML

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6. In what types of matrices is SVD especially useful?

Singular Value Decomposition (SVD) is **especially useful** in a wide range of matrix types — but it shines **particularly in certain scenarios**. Here's a breakdown:

✅ **SVD is Especially Useful For:**

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1. Non-Square Matrices

- 🔷 Works for any m×n matrix (not just square!)
- 🔷 Useful when the matrix is **tall or wide**
- 🔷 Great for **dimensionality reduction** or **low-rank approximation**

Example: user-item rating matrix in recommender systems

2. Low-Rank or Nearly Low-Rank Matrices

- 🔷 If a matrix can be approximated well using just a few singular values
- 🔷 Helps to **compress data** without losing much information
- 🔷 Ideal for **image compression** and **topic modeling**

In practice, many real-world datasets are low-rank or close to it.

3. Noisy or Incomplete Data

- 🔷 SVD helps to **denoise** and **reconstruct missing values**
- 🔷 Often used in **collaborative filtering** (Netflix, YouTube recommendations)
- 🔷 Useful in **signal processing** and **bioinformatics**

4. Text and NLP (Sparse, High-Dimensional Matrices)

- 🔷 Applied to **term-document matrices**
- 🔷 Used in **Latent Semantic Analysis (LSA)** to find hidden topics
- 🔷 Handles **sparsity and high dimensionality** well

5. Correlated or Redundant Features

- 🔷 SVD helps identify directions with the most **variance**
- 🔷 Removes **redundancy** in the feature space
- 🔷 A solid foundation for **PCA** (Principal Component Analysis)

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⚠️ **Less Effective For:**

- Very large, **dense matrices** (may require randomized SVD or other approximations)
- Cases where **interpretability** of components is critical (SVD components are orthogonal but abstract)

🧠 **In Summary:**

Matrix Type	**Why SVD is Useful**
Non-square (m ≠ n)	SVD always works
Low-rank	Allows compression & approximation
Noisy / Missing values	SVD helps reconstruct clean version
Sparse / high-dimensional (NLP)	Great for extracting structure in text
Correlated features	SVD reveals principal directions of variance

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7. Why is SVD more numerically stable than eigen decomposition for PCA?

🧮 **1. What Does PCA Do?**

- PCA finds directions (components) that **maximize variance** in data.
- This is done by computing **eigenvectors of the covariance matrix** $X^T X$ (if XX is zero-centered).

⚠️ **2. What's the Problem with Eigen decomposition?**

- Eigen decomposition requires forming the **covariance matrix** $X^T X$, which:
    - Can **amplify numerical errors**
    - May be **ill-conditioned** if features are highly correlated or data is noisy
    - Increases the chance of **rounding errors**, especially in high dimensions

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✅ 3. Why SVD is Better (More Stable):

Reason	Description
✅ **No covariance matrix needed**	SVD operates directly on the original data matrix XX, avoiding XTXX^T X entirely
✅ **Works on any shape matrix**	Even if XX is not square or is rank-deficient
✅ **Stable with floating point**	Algorithms for SVD are optimized for **floating-point arithmetic**
✅ **Always exists**	SVD is guaranteed to exist for any matrix, eigen-decomposition is not

📐 Geometric View:

- In eigen decomposition, tiny numerical errors in the covariance matrix affect eigenvalues a lot.
- SVD avoids this by decomposing the actual data matrix, so **variability is captured directly**, without squaring the data matrix.

🧠 Summary:

Eigen decomposition	SVD
Uses $X^T X$ or $X X^T$	Uses XX directly
Sensitive to noise & scaling	More robust & stable
Can fail on non-full-rank matrices	Always works
May introduce rounding errors	Less error-prone numerically

✅ So in PCA:

Using **SVD** is numerically more stable and preferred — especially for **high-dimensional**, **noisy**, or **ill-conditioned** data.

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8. What is the rank of a matrix in terms of its singular values?

📏 What Is Matrix Rank (via Singular Values)?

The **rank** of a matrix is the number of **linearly independent rows or columns**.
In terms of **Singular Value Decomposition (SVD)**:

🔢 **The rank = the number of non-zero singular values** in the Σ (Sigma) matrix.

🎯 Formally:

Given the SVD of matrix AA:

$A=U\,\Sigma\,V^T$

- Σ\Sigma is a diagonal matrix with singular values $\sigma1,\sigma2,\ldots,\sigma_r$
- If $\sigma_i > 0$, then that direction contributes to the matrix's structure
- The number of **non-zero σ_i** = **rank of matrix AA**

💡 Example:

If the singular values of a matrix are:

[5.3,3.1,0.0,0.0]

Then:

Rank = 2 (only 2 non-zero singular values)

⚠️ Numerics Note:

Due to floating-point precision, singular values near **zero** (like 1×10^{-12}) are often treated as zero.

So in practice:

np.sum(singular_values > $1e^{-10}$)

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📦 **Summary Table:**

Singular Values (Σ)	Rank
[4.2, 1.0, 0.0]	2
[5.1, 3.0, 2.1]	3
[1e-15, 1.1]	1 (thresholded)

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9. What is the condition number of a matrix, and how does SVD help compute it?

❓ What Is the Condition Number of a Matrix?

The **condition number** of a matrix tells you **how sensitive** a system of linear equations is to small changes or errors in the input.

💡 **Intuition:**

- It measures how stable or **ill-conditioned** a matrix is.
- High condition number = **small changes in input → big changes in output**.
- Low condition number = **more stable**.

🧮 **Definition:**

For a matrix AA, the **condition number** is:

$\text{cond}(A) = \|A\| \cdot \|A^{-1}\|$

For most purposes (especially with SVD), this simplifies to:

$\text{cond}(A) = \sigma_{max} / \sigma_{min}$

Where:

- σ_{max} = **largest singular value**
- σ_{min} = **smallest non-zero singular value**

✅ **How SVD Helps Compute It**

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Given SVD of A:

$A=U\,\Sigma\,V^T$

- You just need to look at the diagonal entries of Σ, which are the **singular values**.
- So, computing the condition number using SVD is both **easy and numerically stable**.

🔢 **Python Example:**

```
import numpy as np


A = np.random.rand(3, 3)

U, S, VT = np.linalg.svd(A)


condition_number = S[0] / S[-1]

print("Condition number:", condition_number)
```

📌 **Interpretation:**

Condition Number	**Interpretation**
~1	Well-conditioned
10–1000	Moderately sensitive
>> 1000	Ill-conditioned
∞	Singular / Non-invertible

🧠 **Summary:**

- **Condition number = ratio of largest to smallest singular value**
- High condition number → **numerical instability**
- SVD makes it easy and **reliable** to compute

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10. Why is SVD defined for all matrices (even non-square ones)?

Because SVD is based on **orthogonal projections** and **factorizations**, not on properties that require the matrix to be square, like **eigenvalues**.

📜 **Quick Recap:**

For any real **m×n** matrix A, SVD is:

$A=U\Sigma V^T$

Where:

- U is m×m (orthogonal matrix)
- Σ is m×n (diagonal matrix with singular values)
- V^T is n×n (orthogonal transpose)

🧠 **Why It Works Universally:**

1. Based on Inner Products, Not Eigenvalues

- SVD decomposes **any real matrix** by looking at the action of AA on input vectors.
- It uses the **eigen-decompositions** of the symmetric matrices ATAA^T A and AATA A^T, which are always **square and positive semi-definite** — so their eigenvalues always exist and are non-negative.

2. No Invertibility Required

- Square matrices may not have eigenvalues (or a full set of linearly independent eigenvectors) unless they're **diagonalizable**.
- SVD doesn't require this — it always gives a full decomposition.

3. Orthogonal Basis for All Matrices

- SVD provides **orthonormal bases** for the column space and row space of any matrix, even if it's **tall, wide, or rank-deficient**.
- That makes it ideal for **dimensionality reduction**, **data compression**, and **low-rank approximations**.

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📦 **Summary:**

Property	Eigen decomposition	SVD
Requires square matrix?	✅ Yes	❌ No
Always defined?	❌ Not always	✅ Always
Works on rectangular data?	❌ No	✅ Yes
Basis for PCA & compression	🚫 Limited	✅ Preferred

🔁 **So in short:**

SVD is defined for all matrices because it works through **symmetric, positive semi-definite matrices** A^TA and AA^T, which always have real, non-negative eigenvalues — enabling the decomposition regardless of shape.

🔷 Application-Based Questions

11. **How is SVD used in dimensionality reduction?**

Explain how you'd reduce a high-dimensional dataset using truncated SVD.

SVD is one of the most powerful tools for **dimensionality reduction**, especially in **PCA**, **latent semantic analysis**, **image compression**, and more.

Let's break it down clearly:

📉 **How SVD is Used in Dimensionality Reduction**

Given a matrix $A \in R\ m \times n$, its **Singular Value Decomposition (SVD)** is:

$A=U\ \Sigma\ V^T$

Where:

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- U and Vare orthogonal matrices (contain the left/right singular vectors)
- Σ is a diagonal matrix of **singular values** (sorted in decreasing order)

🧠 **Core Idea: Low-Rank Approximation**

The most important singular values capture **most of the important structure/variance** in the data.

So, we **keep only the top kk singular values** and corresponding vectors:

$$A_k = U_k \Sigma_k V_k^T$$

Where:

- $U_k \in R^{m \times k}$
- $\Sigma_k \in R^{k \times k}$
- $V_k \in R^{n \times k}$

📦 **Result:**

- A_k is a **low-rank approximation** of A
- It captures the **essential structure** (dominant patterns or variance)
- Reduces data from **n features → k features** (where $k \ll n$)

✏️ **Practical Example in Python:**

```
import numpy as np
from sklearn.decomposition import TruncatedSVD

X = np.random.rand(100, 50)  # 100 samples, 50 features

svd = TruncatedSVD(n_components=10)
X_reduced = svd.fit_transform(X)
```

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```
print(X_reduced.shape)  # Output: (100, 10)
```

Now X_reduced is a 10-dimensional representation of the original 50-D data.

📈 Applications:

Application	What SVD Does
PCA	Extracts top components of variance
Image Compression	Stores only top singular values for reconstruction
Latent Semantic Analysis	Captures semantic structure in text corpora
Noise Reduction	Removes small singular values (likely noise)

📉 Key Insight:

By keeping the top kk singular values, you reduce dimensionality **while preserving most of the data's "energy" or variance**.

12.How is SVD used in recommendation systems (e.g., Netflix or Amazon)?

- Talk about low-rank matrix approximation and collaborative filtering.

Excellent — this is where **SVD** shines in real-world applications like **Netflix**, **Amazon**, and **Spotify** recommendations. Let's break it down step by step 👇

🎯 Goal of a Recommendation System

We want to predict **user preferences** for items they haven't rated yet — e.g., what movie a user would like to watch next.

We usually represent this data as a **user–item matrix**:

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```sql
           Movie A   Movie B   Movie C   ...
User 1       5         ?         3
User 2       ?         4         ?
User 3       2         1         ?
```

- Rows = users
- Columns = items (movies, products)
- Entries = ratings (or interactions)
- Most values are **missing** (sparse matrix)

💡 Enter SVD for Matrix Factorization

We approximate the original **user–item matrix RR** using SVD:

$R \approx U \Sigma V^T$

But in recommendation systems, we often use a **truncated and adapted version**, since RR is incomplete (not all ratings are known).

So instead, we **factor RR** into:

$R \approx PQ^T$

Where:

- $P \in R^{m \times k}$: latent features for users
- $Q \in R^{n \times k}$ latent features for items
- k = number of **latent features** (e.g. genres, styles, user taste)

Each user and item is represented in a **shared latent space**, and the **dot product** of their vectors predicts the rating.

📦 Example:

predicted_rating = np.dot(user_vector, item_vector)

That dot product gives an estimate of how much the user will like that item.

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🛠️ How to Learn PP and QQ?

We minimize the error between known ratings and predicted ones using a loss function:

$$\min_{P,Q} \sum_{(i,j)\in K} (R_{ij} - P_i \cdot Q_j^T)^2 + \lambda(||P_i||^2 + ||Q_j||^2)$$

Where:

- K: set of known ratings
- λ: regularization term

This is often done using **Stochastic Gradient Descent (SGD)** or **Alternating Least Squares (ALS)**.

✅ Benefits of SVD in Recommenders:

Benefit	Why it matters
Latent factor modeling	Captures hidden patterns (e.g., action fan, comedy lover)
Dimensionality reduction	Reduces noise and focuses on key features
Predict missing entries	Suggests unseen movies/products
Scalable	Works well for large sparse matrices

🔁 Used In:

- **Netflix Prize** competition
- **Amazon's item-to-item collaborative filtering**
- Spotify music recommendations
- Goodreads book suggestions

🧠 Summary:

SVD helps break down a massive, sparse user-item interaction matrix into **dense user and item embeddings**, enabling **fast, accurate recommendations**.

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13. **How would you use SVD to compress an image?**

- Describe the steps to convert an image into a low-rank approximation using SVD.

Compressing an image using **SVD (Singular Value Decomposition)** is a classic and intuitive way to understand how SVD works with real-world data like pixel matrices.

Let's walk through it step-by-step 👇

🖼️ 1. Represent the Image as a Matrix

A **grayscale image** can be represented as a 2D matrix where each element is a pixel value (0–255).

A **color image** (RGB) is three such matrices (for Red, Green, and Blue channels).

🔢 2. Apply SVD

Given a grayscale image matrix $A \in \mathbb{R}^{m \times n}$, compute the SVD:

$$A = U\Sigma V^T$$

Then create a **low-rank approximation**:

$$A_k = U_k \Sigma_k V_k^T$$

Where k is the number of singular values/components you keep. Smaller k = more compression.

💡 3. Keep Only Top k Components

By keeping only the **top k** singular values and corresponding vectors:

- You reduce storage from m×n to k(m+n+1)
- You maintain most of the **visual quality**

✅ 4. Reconstruct and Save

Reconstruct the compressed image from Ak, and optionally save or display it.

🖍️ Python Example

By Ayesh Abed

```
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image

# Load grayscale image and convert to numpy array
img = Image.open('your_image.jpg').convert('L')  # L = grayscale
A = np.array(img)

# Perform SVD
U, S, VT = np.linalg.svd(A, full_matrices=False)

# Keep top-k components
k = 50
A_k = np.dot(U[:, :k], np.dot(np.diag(S[:k]), VT[:k, :]))

# Display original vs compressed
plt.subplot(1, 2, 1)
plt.title("Original")
plt.imshow(A, cmap='gray')

plt.subplot(1, 2, 2)
plt.title(f"Compressed (k={k})")
plt.imshow(A_k, cmap='gray')
plt.show()
```

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📦 Storage Compression

Component	Size
Original image	m×n
Compressed SVD	k(m+n+1)

Example: 256×256 image with k=50 Huge reduction!

🧠 Summary:

SVD compresses an image by keeping only the **most significant patterns** (singular values). You lose a little detail, but often retain the **essential structure and appearance** — that's the magic.

14.How does truncating singular values affect the quality of a reconstruction?

This is at the heart of **using SVD for dimensionality reduction and compression**.

🎯 Truncating Singular Values = Reducing Information

When you perform **SVD** on a matrix A:

$$A = U\Sigma V^T$$

You get:

- Σ: a diagonal matrix with **singular values** $\sigma_1, \sigma_2, ..., \sigma_r$
  - Sorted from largest to smallest: $\sigma_1 \geq \sigma_2 \geq ... \geq \sigma_r \geq 0$
  - Each one represents the "importance" (energy/variance) of its corresponding component

✂️ Truncating = Keeping Only Top kk

You keep only the top k singular values and vectors:

$$A_k = U_k\Sigma_k V_k^T$$

This is a **low-rank approximation** of A.

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Effect	Explanation
✅ Small k	High compression, but may lose important detail
✅ Large k	Closer to original, but less compression
📉 Truncation error	Given by the sum of squared discarded singular values:
	$\|A - A_k\|_F^2 = \sum_{i=k+1}^{r} \sigma_i^2$
📷 For images	Less k = more blurry or less detailed; high k = sharper image
🔍 For data	Keeping too few singular values may remove important patterns or variance

💡 **Rule of Thumb:**

$$\text{Energy} = \frac{\sum_{i=1}^{k} \sigma_i^2}{\sum_{i=1}^{r} \sigma_i^2}$$

📊 **Visualization (Conceptually):**

Imagine a plot of singular values:

σ

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└──────────────── k

- The curve drops fast at first (most variance in first few values)

By Ayatullah

- Then it flattens — those later values contribute less

You typically truncate **after the "elbow"** of this curve.

🧠 **Summary:**

Truncating singular values reduces dimensionality and storage, but at the cost of **information loss**. If done wisely (choosing kk properly), it **retains the most important structure** while filtering out noise or redundancy.

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15. How can you use SVD to detect multicollinearity in features?

Using **SVD** to detect **multicollinearity** is a smart and mathematically robust technique. Let's break it down:

🔍 **What is Multicollinearity?**

Multicollinearity occurs when **two or more features** in a dataset are **highly linearly correlated**, which can:

- Cause instability in regression models (e.g., coefficients swing wildly)
- Inflate standard errors
- Reduce model interpretability

🧠 **How SVD Helps**

Suppose you have a feature matrix $X \in \mathbb{R}^{n \times p}$, where:

- n: samples
- p: features

We perform SVD on X:

$$X = U\Sigma V^T$$

Where:

- Σ: diagonal matrix of **singular values** $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$
- These singular values relate to the **variance** in your feature space

Atatullah Abed

🚩 Multicollinearity Clue: Small or Zero Singular Values

If one or more **singular values are near-zero**, it means:

- Some feature dimensions **contribute almost no new information**
- There's a **linear dependency** → **multicollinearity**

📉 Use the Condition Number as a Metric

The **condition number** of XX is:

$$\kappa(X) = \frac{\sigma_{\max}}{\sigma_{\min}}$$

- High κ\kappa (e.g., > 1000) suggests strong multicollinearity
- Infinite κ\kappa: perfect multicollinearity (exact linear dependency)

🛠️ Example in Python

```
import numpy as np
from numpy.linalg import svd

# Example feature matrix
X = np.array([[1, 2, 3],
              [2, 4, 6],
              [3, 6, 9]])  # Perfect multicollinearity

# Perform SVD
U, S, VT = svd(X)

print("Singular values:", S)
print("Condition number:", S[0] / S[-1])
```

Output:

By Ayatuollah Awed

Singular values: [~14.0, ~0, ~0]

Condition number: Very large (or inf)

✅ **Summary**

Feature	What It Means
Small singular values	Feature redundancy / dependency
Condition number ≫1	Strong multicollinearity
Rank-deficient matrix (rank < p)	Perfect multicollinearity

🧠 **Pro Tip:**

SVD is more stable and insightful than a correlation matrix for multicollinearity because it captures **all linear dependencies**, not just pairwise ones.

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16. How is SVD used in natural language processing (e.g., Latent Semantic Analysis)?

SVD plays a powerful role in **Natural Language Processing (NLP)** — especially in **Latent Semantic Analysis (LSA)**, which is one of the earliest and most intuitive uses of SVD for uncovering meaning in text.

🧠 **What is Latent Semantic Analysis (LSA)?**

LSA (or LSI – Latent Semantic Indexing) uses **SVD** to uncover **hidden (latent) semantic structure** in a large corpus of text by reducing its dimensionality.

🧱 **How It Works (Step-by-Step):**

1. Build the Term-Document Matrix

Create a matrix $A \in R_{m \times n}$

- m: number of **terms (words)**

By Aytuollah Abed

- n: number of **documents**
- Entry Aij }: usually **TF-IDF** value of term *i* in document *j*

2. Apply SVD

$A=U\Sigma V^T$

- U: word-topic matrix (terms in reduced semantic space)
- Σ: singular values (importance of each latent topic)
- V^T: document-topic matrix (documents in reduced semantic space)

3. Truncate SVD

Keep only top k singular values and vectors:

$$A_k=U_k\,\Sigma_k\,V_k^T$$

Now:

- Each **word** is a vector in a lower-dimensional space (semantic meaning)
- Each **document** is also a vector in this space

This removes noise and reveals **semantic relationships** (e.g., synonyms, related concepts)

🔍 **What It Solves**

Problem	**LSA/SVD Solution**
Synonymy (different words, same meaning)	Maps to same latent concept
Polysemy (same word, multiple meanings)	Places word in context of documents
High dimensionality	Reduces to core semantic dimensions

🧪 **Example: Tiny LSA Workflow**

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.decomposition import TruncatedSVD

docs = ["I like deep learning.",
```

By Ayatullah Abed

```
    "Deep learning is fun.",
    "NLP includes deep learning techniques."]

# TF-IDF matrix
vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(docs)

# Apply SVD (LSA)
svd = TruncatedSVD(n_components=2)
X_reduced = svd.fit_transform(X)

print("Reduced vectors (semantic space):\n", X_reduced)
```

🧠 **Summary:**

SVD Component	**Meaning in NLP (LSA)**
U_k	Word vectors in semantic space
Σ_k	Strength of latent semantic concepts (topics)
V_k^T	Document vectors in semantic space

Latent Semantic Analysis = SVD applied to text data to uncover the **meaning structure** hidden in word-document relationships.

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17. **When would you prefer using Truncated SVD over PCA?**

While **PCA** and **Truncated SVD** are both used for dimensionality reduction, they have **key differences** — and the choice between them depends largely on your **data type and format**.

🔍 **When to Prefer Truncated SVD over PCA**

✅ **1. Sparse or Large-Scale Data (e.g., text data)**

By Ayatollah Abed

- **Truncated SVD** works **directly on sparse matrices** (like TF-IDF from text), without converting them to dense format.
- **PCA**, as implemented in many libraries (like sklearn), requires **centering** the data, which can make a sparse matrix **dense** → memory explosion ⚠️

🔧 **Use case:** NLP, document-term matrices, recommendation systems.

✅ 2. You Don't Need Mean-Centering

- **PCA** subtracts the mean from data before decomposing, which can change the interpretation or structure of the data.
- **Truncated SVD** does **not center** the data (by default), which is fine (and even preferred) for certain applications like **LSA (Latent Semantic Analysis)**.

✅ 3. Performance/Scalability

- Truncated SVD (e.g., sklearn.decomposition.TruncatedSVD) is **faster** and more **memory-efficient** for **high-dimensional, sparse datasets**.
- PCA becomes slow and memory-intensive with large datasets.

✅ 4. Dimensionality Reduction for Non-centered Data

In situations where **centering is not desired** or meaningful (e.g., for non-negative matrices or count data), **Truncated SVD** is a safer choice.

🆚 Summary Table: PCA vs Truncated SVD

Feature	PCA	Truncated SVD
Requires centering?	✅ Yes	❌ No
Works with sparse data?	❌ Not directly	✅ Yes
Use case	Numerical, dense datasets	Text data (TF-IDF), sparse features

By Aya Tullah Abed

Feature	PCA	Truncated SVD
Output vectors	Orthogonal principal components	Latent features (not necessarily orthogonal)
Interpretation	Maximize explained variance	Approximate decomposition, lower-rank

💡 **In Practice**

Use **Truncated SVD** when:

- You're working with **text features** (TF-IDF, count vectors)
- Your matrix is **large and sparse**
- You're applying **LSA**
- You want a **fast, memory-efficient** method

Use **PCA** when:

- Your data is **dense and numerical**
- You care about **maximizing explained variance**
- You want to visualize components in 2D/3D

🔷 Implementation & Coding Questions

18.How do you compute SVD in Python using NumPy or SciPy?

Computing **Singular Value Decomposition (SVD)** in Python is straightforward using either **NumPy** or **SciPy**. Here's how you can do it with both:

By Ayatullah Abed

✅ 1. Using NumPy

```python
import numpy as np

# Example matrix
A = np.array([[1, 2], [3, 4], [5, 6]])

# Compute SVD
U, S, VT = np.linalg.svd(A, full_matrices=False)

print("U:\n", U)
print("Singular values (S):\n", S)
print("V^T:\n", VT)
```

Output:

- U: Left singular vectors
- S: Array of singular values
- VT: Right singular vectors (transposed)

Note: To reconstruct the matrix:

```python
S_matrix = np.diag(S)
A_reconstructed = U @ S_matrix @ VT
```

✅ 2. Using SciPy

By Ayman Abdullah Awad

```python
from scipy.linalg import svd

# Same matrix
A = np.array([[1, 2], [3, 4], [5, 6]])

# Compute SVD
U, S, VT = svd(A, full_matrices=False)

print("U:\n", U)
print("Singular values:\n", S)
print("V^T:\n", VT)
```

The SciPy version is often more numerically robust and offers more control for advanced applications (like computing only part of the decomposition).

⚡ **Quick Notes:**

Component	Shape	Meaning
U	(m × k)	Left singular vectors (columns = orthonormal basis of column space)
S	(k,)	Singular values (square roots of eigenvalues of A^TA)
V^T	(k × n)	Right singular vectors transposed

- k=min(m,n)
- S is **not** a matrix in NumPy/SciPy by default — it's a **1D array**

19.How do you reconstruct the original matrix from its SVD components?

Reconstructing the original matrix from its **SVD components** is straightforward using this formula:

$A=U\cdot\Sigma\cdot V^T$

Where:

- U: left singular vectors

By Ayatullah Ayed

- Σ: diagonal matrix of singular values
- V^T: transpose of right singular vectors

🧪 Example in Python:

```python
import numpy as np

# Original matrix
A = np.array([[1, 2], [3, 4], [5, 6]])

# Compute SVD
U, S, VT = np.linalg.svd(A, full_matrices=False)

# Convert singular values to diagonal matrix
Sigma = np.diag(S)

# Reconstruct the original matrix
A_reconstructed = U @ Sigma @ VT

print("Reconstructed A:\n", A_reconstructed)
```

🔁 Explanation:

- np.linalg.svd() gives you:
    - U: shape (m, k)
    - S: 1D array of k singular values
    - VT: shape (k, n)
- np.diag(S) turns the 1D array S into a square diagonal matrix Σ\Sigma
- Matrix multiplication (@) reassembles the original matrix

✅ The result should closely match the original A, with only very small differences due to floating-point precision.

🎯 Optional: Low-Rank Approximation (Truncated SVD)

You can also reconstruct a **compressed version** by keeping only the top rr singular values:

By Avatar

```python
r = 1  # number of singular values to keep
U_r = U[:, :r]
S_r = np.diag(S[:r])
VT_r = VT[:r, :]

A_approx = U_r @ S_r @ VT_r
```

This is useful for **dimensionality reduction** or **compression**.

20.Write a function to compress an image using only the top k singular values.

Here's a Python function to compress a grayscale image using **Singular Value Decomposition (SVD)** and retaining only the **top k singular values**. This method works well for image compression by reducing the dimensionality of the image matrix.

If your image is in **color (RGB)**, you can apply this separately to each color channel.

✅ Code (Using NumPy and Matplotlib):

```
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image

def compress_image_svd(image_path, k):
    # Load the image and convert to grayscale
    original_img = Image.open(image_path).convert('L')  # 'L' mode is grayscale
    A = np.array(original_img, dtype=np.float64)

    # Apply SVD
    U, S, VT = np.linalg.svd(A, full_matrices=False)
```

By Ayatollah Abed

```
# Keep only top k singular values
S_k = np.diag(S[:k])
U_k = U[:, :k]
VT_k = VT[:k, :]

# Reconstruct the compressed image
A_k = np.dot(U_k, np.dot(S_k, VT_k))

# Clip values to valid range and convert to uint8
A_k = np.clip(A_k, 0, 255)
compressed_img = Image.fromarray(A_k.astype('uint8'))

# Display original and compressed image
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.title("Original Image")
plt.imshow(original_img, cmap='gray')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.title(f"Compressed Image (k={k})")
plt.imshow(compressed_img, cmap='gray')
plt.axis('off')
plt.show()

return compressed_img
```

By Ayatuollah Abed

🧪 **Example Usage:**

compressed = compress_image_svd("example_image.jpg", k=50)

compressed.save("compressed_image_k50.jpg")

📝 **Notes:**

- Larger k values retain more information but give less compression.
- For color images, you can compress each RGB channel independently.

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21. Use SVD to perform low-rank matrix approximation in a collaborative filtering example.

Let's walk through how to **use SVD for low-rank matrix approximation** in a **collaborative filtering** setting — a common technique used in **recommendation systems** (e.g., for predicting movie ratings).

✅ **Goal**

Given a **user-item rating matrix**, use **SVD** to approximate it and **predict missing ratings** by reducing the matrix to a low-rank approximation.

🧠 **Basic Concept**

Suppose we have a matrix R where:

- Rows = users
- Columns = items
- R[i][j] = rating user i gave to item j (or 0/NaN if not rated)

We decompose R using **SVD**:

$$R \approx U_k \cdot S_k \cdot V_k^T$$

By Ayatuollah Abed

This gives a rank-k approximation that captures **latent factors** (e.g., genre preference, item popularity).

🧪 Python Example

```
import numpy as np
import pandas as pd
from numpy.linalg import svd

def svd_collaborative_filtering(ratings_matrix, k):
    # Replace missing ratings (zeros or NaNs) with the user mean for better approximation
    R = ratings_matrix.copy().astype(float)
    user_means = np.true_divide(R.sum(1), (R != 0).sum(1))
    for i in range(R.shape[0]):
        R[i, R[i] == 0] = user_means[i]

    # Apply SVD
    U, S, VT = svd(R, full_matrices=False)

    # Reduce to rank k
    U_k = U[:, :k]
    S_k = np.diag(S[:k])
    VT_k = VT[:k, :]

    # Low-rank approximation
    R_approx = np.dot(U_k, np.dot(S_k, VT_k))

    return R_approx
```

By Ayatuollah Abed

🧪 **Example Usage:**

```
ratings = np.array([
   [5, 3, 0, 1],
   [4, 0, 0, 1],
   [1, 1, 0, 5],
   [0, 0, 5, 4],
   [0, 1, 5, 4]
])

predicted_ratings = svd_collaborative_filtering(ratings, k=2)
print(np.round(predicted_ratings, 2))
```

🔍 **What This Does:**

- Fills in the missing values with estimated ratings.
- Captures underlying user/item patterns (e.g., “users who like sci-fi also like fantasy”).

تم بحمد الله

By Ayatullah Abed